BT1010 Quiz_17-11-2025_AS

Dear Students,

This quiz has 25 MCQs, each for 1 point. You have 50 minutes to submit the quiz. Please submit the quiz by 10.50 am.

Any kind of malpractice during the quiz will not be tolerated.

All the best,

Gunjan and Savita

ic25btech11003@iith.ac.in Switch account

Draft saved

* Indicates required question

Email *

Record **ic25btech11003@iith.ac.in** as the email to be included with my response

What is the process of metastasis in cancer? 1 point

- ○ The shrinking of a tumor due to immune response.
- ○ The uncontrolled division of normal cells.
- ◉ The spread of cancer cells from the primary tumor to distant parts of the body.
- ○ The formation of new blood vessels to supply the tumor.

Clear selection

Which of the following is considered a micronutrient essential for human health? 1 point

- ○ Fats
- ◉ Vitamin C
- ○ Proteins
- ○ Carbohydrates

Clear selection

Which of the following statements is **true** about the structure of an antibody? 1 point

- ◉ Antibodies consist of two identical heavy chains and two identical light chains.
- ○ Antibodies are composed only of heavy chains, with no light chains.
- ○ The variable regions of the antibody are identical across all antibodies.
- ○ Antibodies have a single constant region and a single variable region.

Clear selection

Which option is incorrect: Partial hydrogenation of cooking oil converts 1 point

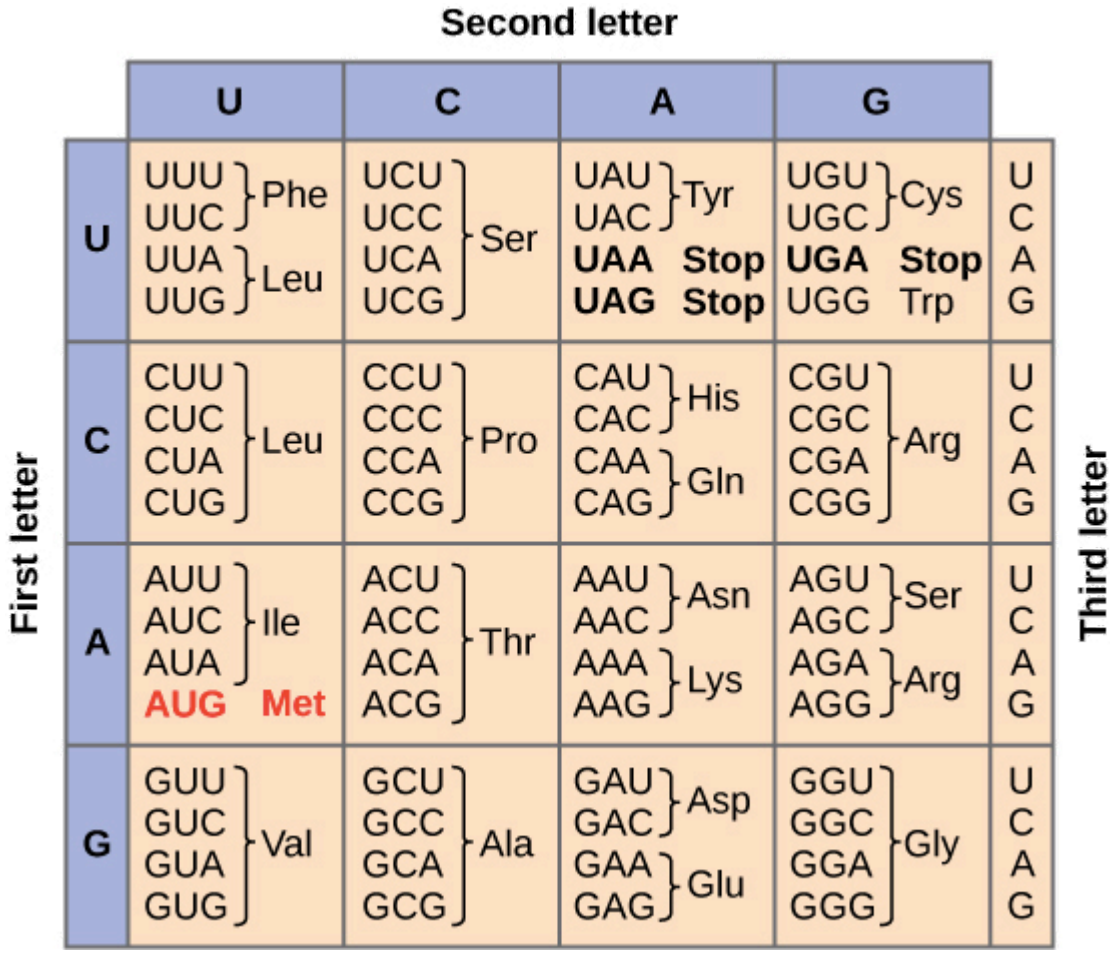
- ○ double bonds of the fatty acids to single bonds
- ○ unsaturated fatty acids to saturated fatty acids
- ◉ cis double bonds to trans double bonds
- ○ triacylgrycerol to fatty acids
- ○ all of the above

Clear selection

Transcribe and translate the following DNA sequence into a peptide sequence: 1 point

Template strand of DNA: 3' GCACCTATGTGAACTAAA 5'

Only write the peptide sequence in this manner for auto correction: (e.g. Arg-Met-Leu-Lys-Trp)



Ala-Pro-Met

Which of the following is **not** a mode of transmission for AIDS? 1 point

- ○ Sharing needles with an infected person
- ◉ Casual contact, such as hugging or shaking hands
- ○ Unprotected sexual contact with an infected person
- ○ Through blood transfusions from an infected donor

Clear selection

An enzyme with lower Km can easily be saturated with the substrate compared to an enzyme with higher Km. 1 point

- ○ True
- ○ False

Ribosomes are made up of 1 point

- ○ carbohydrates, proteins and rRNAs
- ○ plasma membrane, proteins and rRNAs
- ○ rRNAs, proteins and tRNAs
- ○ rRNAs and proteins

Why are T cells called "T cells"? 1 point

- ○ They are primarily involved in the Transport of immune cells.
- ◉ Because, they mature in the Thymus gland.
- ○ They have a unique "T" shaped receptor.
- ○ Because, they are named after their role in the production of Tumor necrosis factor (TNF).

Clear selection

Which of the following organelle/reaction produces glucose? 1 point

- ○ mitochondria
- ○ calvin cycle
- ○ light reaction
- ○ thylakoid
- ◉ all of the above

Clear selection

Which of the following cell organelle is absent in prokaryotic (bacterial) cells? 1 point

- ◉ ribosomes
- ○ cell wall
- ○ mitochondria
- ○ flagella
- ○ cytoplasm
- ○ none of the above

Clear selection

Structure of the DNA was revealed using 1 point

- ○ Agarose Gel Electrophoresis
- ◉ X-Ray Diffraction
- ○ Fluorescence In Situ Hybridization (FISH)
- ○ Transformation
- ○ Cloning

Clear selection

Which of the following cells are responsible for producing antibodies? 1 point

- ◉ T cells
- ○ B cells
- ○ Macrophages
- ○ Neutrophils

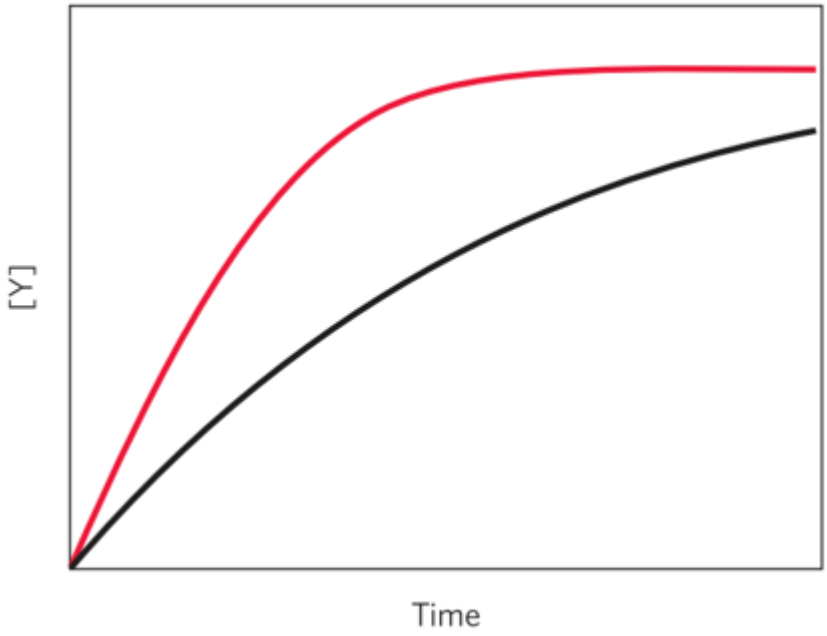
Clear selection

Polymerase Chain Reaction (PCR) is used to 1 point

- ◉ amplify specific region of DNA
- ○ transcribe DNA to RNA
- ○ purify DNA
- ○ separate DNA

Clear selection

An enzyme that catalyzes the reaction X to Y is isolated from two bacterial species. The enzymes have the same Vmax, but different Km values for the substrate X. Enzyme A has a Km of 2.0 μM, while enzyme B has a Km of 0.5 μM. The plot below shows the kinetics of reactions carried out with the same concentration of each enzyme and with [X]= 1 μM. Which curve corresponds to which enzyme? 1 point



- ○ Red curve represents Enzyme A
- ◉ Black curve represents Enzyme A

Clear selection

Which of the following is a characteristic of innate immunity? 1 point

- ◉ It provides long-lasting immunity after an infection.
- ○ It includes physical barriers such as skin and mucous membranes.
- ○ It is specific to particular pathogens.
- ○ It is activated only after exposure to a pathogen.

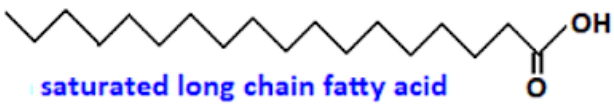
Clear selection

Following gel matrix is used to separate proteins by electrophoresis. 1 point

- ○ Agarose gel
- ○ Ethidium bromide
- ◉ Polyacrylamide gel
- ○ Gelatin

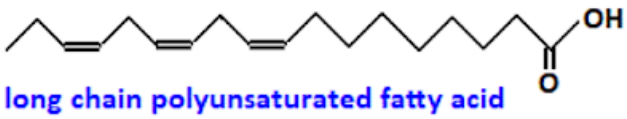
Clear selection

Which of the following fatty acids has a higher melting temperature? 1 point



stearic acid

$C_{17}H_{35}COOH$



linolenic acid (omega 3)

$C_{17}H_{29}COOH$

- ○ Linolenic acid
- ◉ Stearic acid

Clear selection

Which of the following pathogens is responsible for causing Typhoid fever? 1 point

- ○ Streptococcus pneumoniae
- ○ Escherichia coli
- ○ Mycobacterium tuberculosis
- ◉ Salmonella typhi

Clear selection

Your Roll Number *

IC25BTECH11003

What is a sarcoma? 1 point

- ○ A cancer that develops from epithelial cells.
- ○ A viral infection affecting the immune system.
- ○ A type of cancer that originates in connective tissues like bones, muscles, or cartilage.
- ◉ A type of cancer that originates in the skin.

Clear selection

Which of the following best describes *passive immunity*? 1 point

- ○ Immunity that lasts for a lifetime due to memory cell formation
- ○ Immunity that develops after exposure to a pathogen and the body produces its own antibodies
- ○ Immunity acquired when ready-made antibodies are transferred from another individual
- ◉ Immunity obtained by vaccination with weakened or killed pathogens

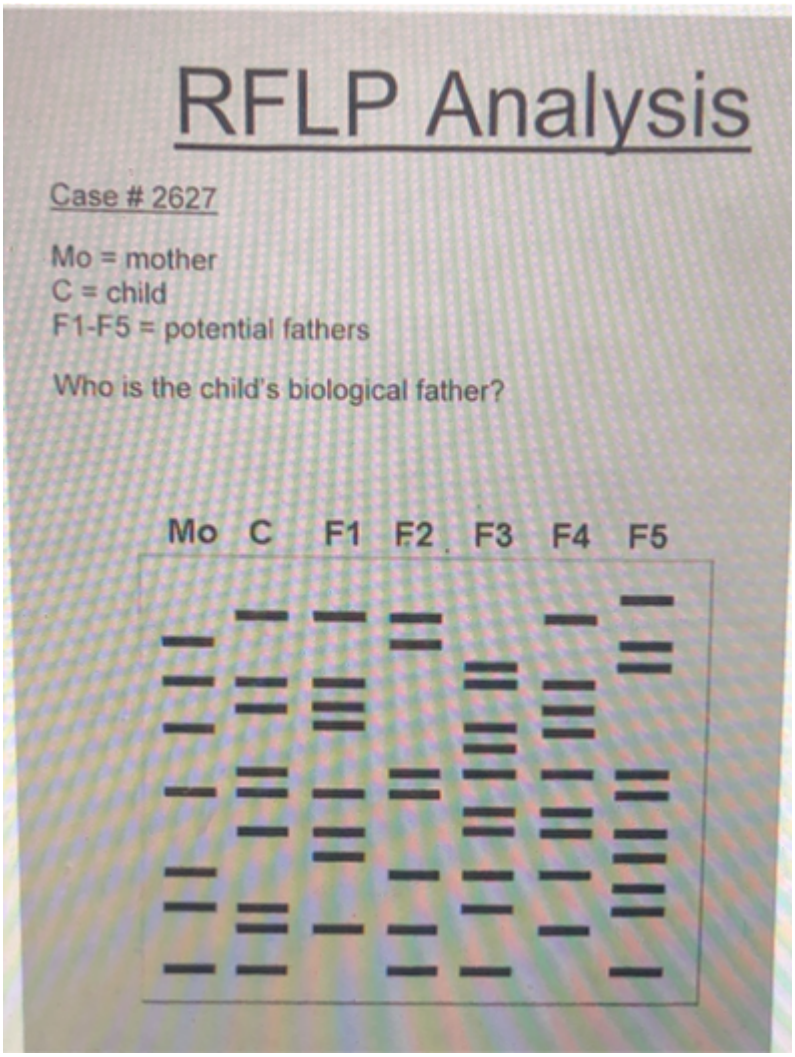
Clear selection

Which of the following is an example of Physiological barrier? 1 point

- (●) Macrophage
- ( ) Saliva
- ( ) Mucus
- ( ) Skin

Clear selection

From this RFLP analysis, who is the biological father? 1 point



- [ ] F1
- [ ] F2
- [ ] F3
- [x] F4
- [ ] F5

Clear selection

D-Glucose and D-Galactose are 1 point

- ○ Disaccharides
- ○ Enantiomers
- ○ Anomers
- ◉ Epimers

Clear selection

Which of the following antibodies is the heaviest? 1 point

- ○ IgE
- ◉ IgM
- ○ IgG
- ○ IgA

Clear selection

A copy of your responses will be emailed to ic25btech11003@iith.ac.in.

Submit Page 1 of 1 Clear form

Never submit passwords through Google Forms.

This form was created inside of IIT Hyderabad. - Contact form owner

Does this form look suspicious? Report

Google Forms